Prince Agyei Tuffour

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SUMMARY

AI scientist with over 2 years of experience designing and deploying machine learning models in e-commerce and academic research settings. Skilled in Python, TensorFlow, PyTorch, and scikit-learn with expertise in recommendation systems, supervised learning, dimensionality reduction, and MLOps practices. Delivered personalized recommendation solutions driving a 15% engagement lift and achieved 92% accuracy in predictive research experiments.

EDUCATION

Oregon State University, Corvallis OR
Master of Science, Mathematics

Kwame Nkrumah University of Science and Technology, Ghana
Bachelor of Science, Mathematics

SKILLS

- **Programming Languages**: Python, Javascript, C++, SQL, Java, Typescript
- **Machine Learning & AI**: Machine Learning, Deep Learning, Keras, ScikitLearn, PyTorch, Tensorflow, Regression, Classification, Natural Language Processing, Large Language Models, Recommender Systems, Clustering, Anomaly Detection, MLOps, YOLO
- **Tools & Technologies**: Git, Nextjs, Redis, AWS, Google Cloud, Playwright, Tailwind CSS, Linux/Unix, Selenium, Flask, Django, RESTful APIs, Express, PostgreSQL, MongoDB, React, SciPy, Pandas, Streamlit, Distributed Computing Frameworks

WORK EXPERIENCE

dynaConnections Corporations **Mar 2025 - Present**
Software QA Engineer

- Design and execute comprehensive test strategies including manual and automated testing protocols, API testing, user interface validation, performance testing, and integration of CI/CD pipelines to automate deployments and monitor test anomalies.
- Achieved 40% improvement in overall product reliability and 60% reduction in post-deployment bugs through implementing quality assurance best practices, establishing anomaly detection mechanisms within test frameworks, and catching edge cases early.
- Developed expertise in creating detailed test cases, performing regression testing and analytical problem-solving that provides deep insights into software development lifecycle and ensures high standards of user experience.

Akomapa Health **Mar 2025 - Present**
Lead Software Developer/Executive Board Member

- Manage complete website workflow including server maintenance, SQL database optimization, and feature implementation that supports 500+ active student users monthly with 99.8% uptime reliability for health education platform.
- Developed automated content delivery systems and mobile-first design improvements incorporating recommender systems to personalize content, increasing user engagement by 65% and boosting mobile traffic by 80%, making health information more accessible to student communities.
- Built interactive community platform enabling students to access peer-reviewed health content, participate in wellness challenges, and connect with certified health professionals, demonstrating how technology amplifies grassroots health advocacy movements.

Cita Marketplace.com **Jun 2022 - 2023**
Machine Learning Engineer - Intern

- Developed and deployed a machine learning recommendation system for the Cita Marketplace platform, enhancing personalized product recommendations and driving a 15% increase in user engagement and conversion rates.
- Analyzed and optimized customer behavior data to create predictive models that improved the accuracy of search and product relevance by 20%, resulting in better customer experiences and retention.
- Collaborated cross-functionally with engineering and product teams to integrate ML solutions seamlessly, accelerating development timelines and contributing to a 10% faster feature rollout.

Oregon State University **Sep 2021 - Dec 2023**
Graduate Research Assistant

- Developed and fine-tuned machine learning models utilizing support vector machines (SVM) and decision trees for predictive analysis of complex data sets, achieving a 92% model accuracy in identifying patterns within mathematical simulations.
- Implemented numerical optimization techniques such as gradient descent and stochastic optimization to solve high-dimensional mathematical problems, leading to a 25% improvement in computational speed.
- Applied principal component analysis (PCA) and t-SNE for dimensionality reduction and visualization of multidimensional data, enhancing the interpretability of model outputs and revealing significant clustering in research data.
- Wrote Python scripts leveraging NumPy, SciPy, and TensorFlow to automate data preprocessing and build machine learning pipelines, which reduced data preparation time by 40% and streamlined model training.

PROJECTS

Guidr App

- Designed and deployed an AI-powered graduate school assistant on AWS using Python and Docker, integrating OpenAI's GPT-4 API to streamline school selection, application prep, and funding search for 1,500+ users
- Built an end-to-end MLOps pipeline with CI/CD in Git and used secure chat storage, web scraping, and LLM-based Q&A to deliver personalized insights, boosting applicant efficiency and success rates

Stormlog - Open Source | https://stormlog.dev

- Contributed to GPU memory profiling tool optimizing deep learning workflows by visualizing model memory usage for TensorFlow and PyTorch.
- Enhanced debugging, training efficiency, and memory usage visualization by identifying bottlenecks, aiding various ML practitioners in improving model performance.

Akomapa Health | https://akomapa.org

- Engineered a Yale- and UCLA-backed health platform using React, Supabase, and TypeScript, reaching 25,000+ users and improving health literacy by 65% in rural Ghana.
- Achieved 80% rise in preventive care in high blood pressure and 300% outreach growth via offline features, volunteer medical training, and integrated emergency contact systems.

Veridex - RAG Evaluation Framework

- Built an end-to-end RAG pipeline with document chunking, embedding generation (OpenAI/Sentence-Transformers), and retrieval via FAISS and
  ChromaDB vector stores, enabling context-grounded Q&A over domain-specific corpora.
- Implemented an LLM evaluation suite measuring faithfulness, answer relevance, and context precision to detect hallucination, achieving 92%+
  grounding accuracy on benchmark datasets using automated scoring and human-in-the-loop validation.